



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2017

BIOLOGY

9744/01

HIGHER 2

25 August 2017

Paper 1 Multiple Choice

1 hour

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, centre number and index number on the Answer Sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

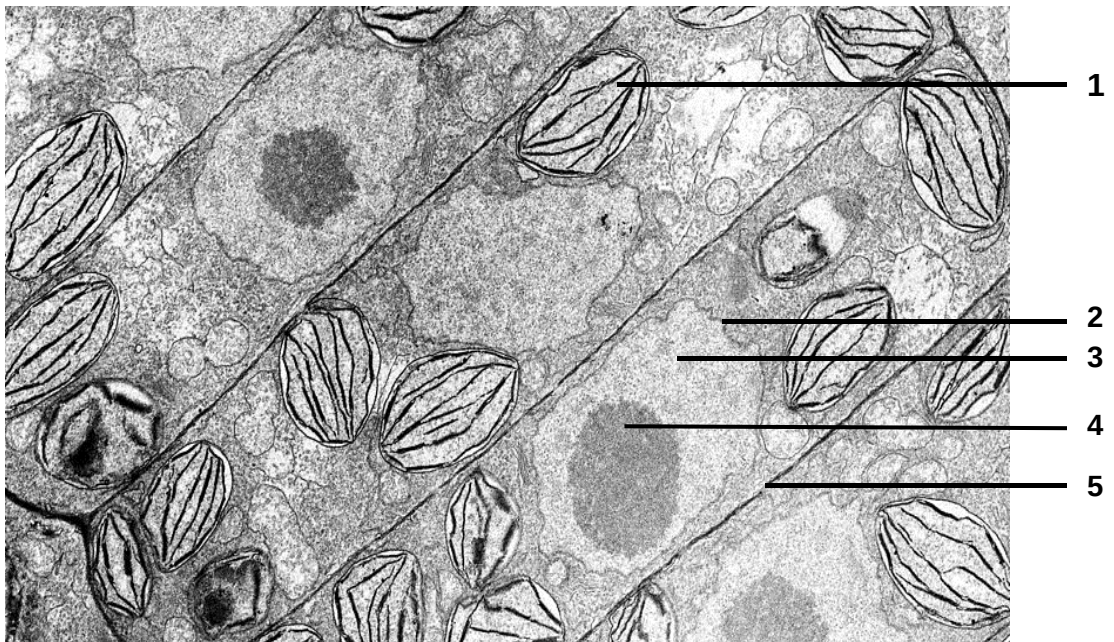
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of **23** printed pages.

- 1 The electron micrograph shows root cells from the duckweed plant.



Which of the following options about structures 1 to 5 is correct?

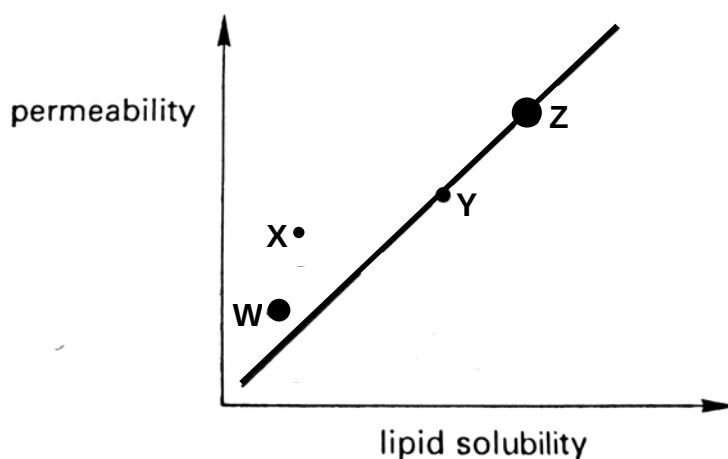
	Contain ribosomal subunits	Contain tRNA	Contain phospholipids
A	1, 2, 3	3 only	1, 4
B	1, 3, 4	1, 3	2, 5
C	1, 3, 5	1, 2, 4	2, 4, 5
D	2, 4, 5	2, 4, 5	1, 3, 4, 5

- 2 Plants are able to regulate their thylakoid membrane fluidity at different seasons of the year. In an investigation on thylakoid membrane fluidity in spinach leaves, three variables which influence membrane fluidity were measured at winter and summer.

Variable	Season X	Season Y
Percentage of saturated fatty acids	15.5	13.9
Average number of double carbon bonds per lipid	4.71	4.76
Lipid to chlorophyll ratio	2.9	2.1

Which of the following correctly identifies season X, with the most possible explanation?

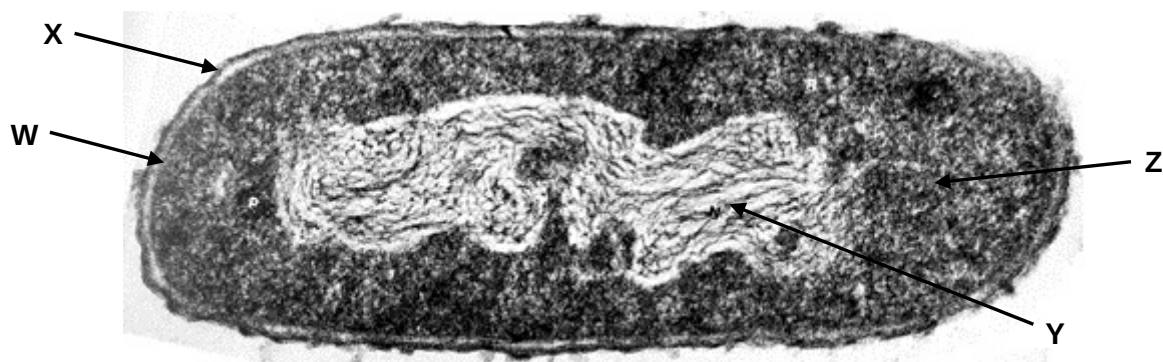
- A** Winter; a higher lipid to chlorophyll ratio increases proportion of weak hydrophobic interactions resulting in a more fluid membrane at lower temperatures
- B** Summer; a higher proportion of saturated fatty acids prevents phospholipids from moving too far apart at higher temperatures
- C** Winter; a higher proportion of saturated fatty acids prevents phospholipids from packing too closely at lower temperatures
- D** Summer; a higher number of kinks per lipid allows phospholipids to pack closely together at higher temperatures
- 3 The diagram shows the relationship between the size, lipid solubility and ability of molecules to cross the mammalian cell surface membrane. The diameter of the black circles in the diagram is proportional to the size of the molecules.



Which of the following could molecules W to Z represent?

	W	X	Y	Z
A	calcium chloride	methane	cholesterol	glucose
B	glucose	water	carbon dioxide	cholesterol
C	calcium chloride	water	glucose	cholesterol
D	glucose	methane	carbon dioxide	calcium chloride

- 6 The diagram shows an electron micrograph of a bacterial cell.



Which of the following correctly identifies the functions of structures W, X, Y and Z?

	W	X	Y	Z
A	Maintains shape of bacterial cell	Protects bacterial cell against desiccation	Contains antibiotic resistance genes which may be beneficial to the bacterial cell	Serves as the site of protein synthesis
B	Controls the passage of substances into and out of the cell	Maintains shape of bacterial cell	Contains genetic information which is essential to the survival of bacterial cell	Serves as the site of translation of mRNA
C	Controls the passage of substances into and out of the cell	Maintains shape of bacterial cell	Contains antibiotic resistance genes which may be beneficial to the bacterial cell	Protects bacterial cell against desiccation
D	Protects bacterial cell against desiccation	Protects bacterial cell from the action of phagocytes	Contains genetic information which is essential to the survival of bacterial cell	Maintains shape of bacterial cell

- 7 The diagram below shows the enterobacteria phage P22 which is a bacteriophage that infects the bacterium *Salmonella typhimurium*. The genetic material and replication cycle of P22 are similar to the lambda phage.

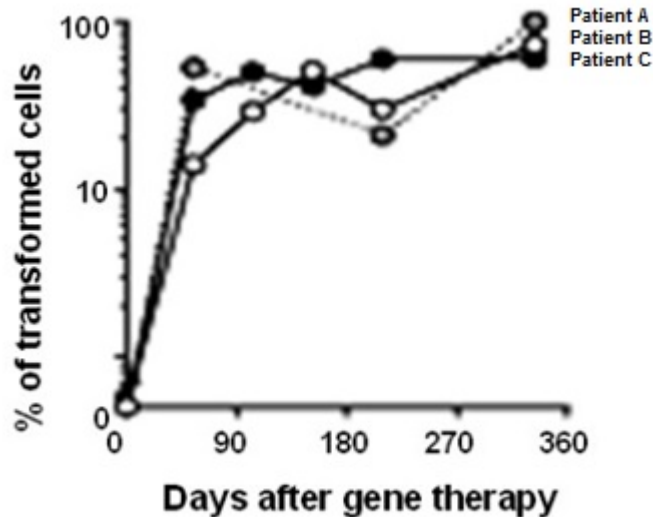


Which of the following statements can be inferred?

- A** It is a virulent phage which contains double-stranded RNA and undergoes the lytic cycle.
- B** It has a spherical envelope that is obtained when the phage buds from the host cell.
- C** It forms a prophage, which is replicated and passed to the daughter bacterial cells during cell division.
- D** It has tail fibres which allow the phage to attach to various species of host bacteria.

- 8 Gene therapy is a technique for correcting defective genes responsible for disease development. It involves introducing a copy of a normal functional gene into target cells with non-functional genes. A vector is a vehicle usually required to deliver the functional allele into the patient's target cells.

Gene therapy was conducted on three patients to treat a neurological disease. The percentage of transformed stem cells introduced into their brains was monitored over a period of 360 days after the gene therapy. However, there was no improvement in their neurological condition.



Which of the following would be the most appropriate investigation to carry out in view of the unsuccessful clinical trial?

- A Regulation of neurone-specific genes in transformed stem cells
- B Effectiveness of vector to deliver the gene into stem cells
- C Modification of stem cells to reduce immune reaction to transformed stem cells
- D Activation of telomerase gene in transformed stem cells

- 9 Many of the most effective antibiotics used in modern medicine are compounds made by fungi that inhibit bacterial protein synthesis. Among the most commonly used drugs are Chloramphenicol, Cycloheximide and Rifampicin. The results of the exposure to eukaryotic and prokaryotic cells to the above three drugs are shown.

Anti-microbial drug	Chloramphenicol	Cycloheximide	Rifampicin
Eukaryotic Animal Cell	Truncated polypeptides were found in mitochondria only	Truncated polypeptides were found in cytosol	No protein synthesized
Prokaryotic Cell	Truncated polypeptides were found in the cytosol	Truncated polypeptides were found in the cytosol	No protein synthesized

Which of the following shows the correct combination of the possible drug mechanisms of the above drugs?

	Chloramphenicol	Cycloheximide	Rifampicin
A	Inhibits the peptidyl transferase activity of the 70S ribosomes	Inhibits elongation by binding the E site of the ribosome hence preventing the release of tRNA	Inhibits the transcription of DNA by blocking the movement of RNA polymerase on DNA
B	Inhibits the peptidyl transferase activity of the 80S ribosomes	Inhibits elongation by binding the E site of the ribosome hence preventing the release of tRNA	Inhibits the transcription of DNA by blocking the movement of RNA polymerase on DNA
C	Inhibits elongation by binding the P site of the ribosome hence preventing the formation of peptidyl tRNA	Inhibits elongation by binding the A site of the ribosome hence preventing the release of tRNA	Inhibits translation by binding to the small ribosomal subunit
D	Inhibits elongation by binding to mRNA and preventing ribosomal translocation	Inhibits elongation by binding the P site of the ribosome hence preventing the release of polypeptide	Inhibits translation by binding to the binding site of large ribosomal subunit

10 The list shows the stages in the cellular replication of DNA.

- 1 Formation of phosphodiester bonds between Okazaki fragments
- 2 Dissociation of DNA from histone proteins
- 3 Synthesis of RNA primers
- 4 Addition of deoxyribonucleotides
- 5 Separation of DNA double helix

Which is the correct sequence?

- A** 5 → 4 → 3 → 1 → 2
- B** 2 → 5 → 4 → 3 → 1
- C** 5 → 2 → 3 → 1 → 4
- D** 2 → 5 → 3 → 4 → 1

11 Proteins X and Y play a role in regulating gene expression.

Protein X forms a complex with GTP and mediates the binding of methionyl aminoacyl-tRNA to the small ribosomal subunit which then binds to the 5' end of mRNA and scans for the first AUG codon. When an AUG codon is recognised, protein X hydrolyses bound GTP to GDP and it is released from the small ribosomal subunit. A complete ribosome then forms and protein synthesis begins.

Protein Y is required to cause GDP release from protein X so that it can be reused. The reuse of protein X is inhibited when it is phosphorylated because phosphorylated X binds to protein Y tightly and inactivates protein Y.

Which of the following statements can be concluded?

- 1 Proteins X and Y control gene expression by regulating translational control.
- 2 Protein X is a translational initiation factor that has catalytic activity.
- 3 Inactivation of protein Y will inhibit translation.
- 4 Active protein kinases will decrease overall protein synthesis.

- A** 1, 2, 3 and 4
- B** 1 and 2 only
- C** 1 and 3 only
- D** 2 and 3 only

- 12** Which of the following correctly matches the state of the *lac* operon to the presence/absence of the molecule(s)?

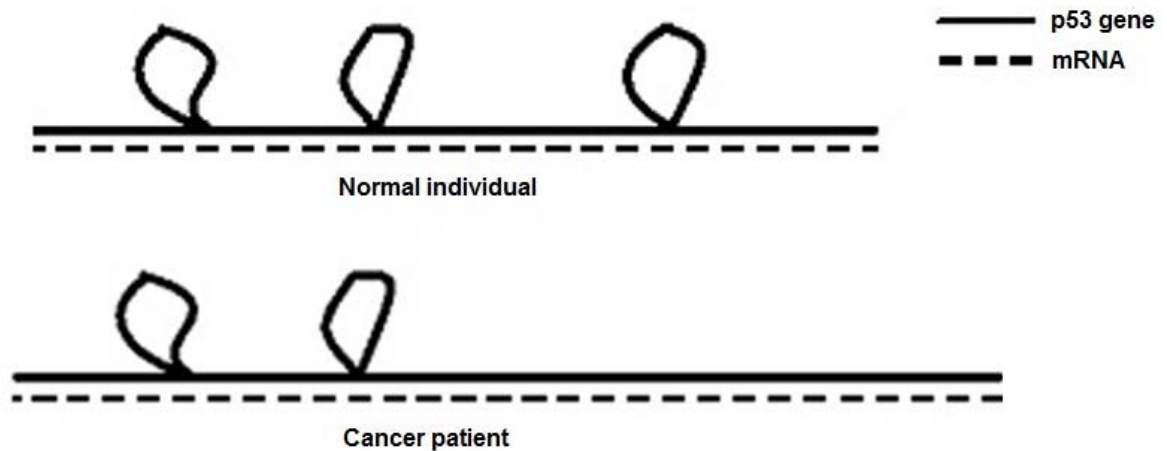
	State of <i>lac</i> operon	Glucose	Lactose	cAMP
1	On	Present	Present	Absent
2	On	Absent	Present	Absent
3	Off	Present	Absent	Absent
4	On	Absent	Present	Present

- A** 1, 2, 3 and 4
B 1, 3 and 4 only
C 1 and 2 only
D 2 and 3 only

- 13** Which of the following correctly matches the step involved in Southern blotting to its purpose?

	Step	Purpose
A	Transferring DNA fragments from a gel to a nitrocellulose paper with the use of alkaline solution	To permanently attach the DNA fragments to a surface and to separate the two complementary DNA strands
B	Adding a radioactive probe	To bind to all DNA fragments for visualisation of the DNA bands
C	Adding restriction enzymes	To digest each DNA sample into fragments of similar lengths
D	Electrophoresis	To separate DNA fragments based on the amount of negative charges the fragment has

- 14 mRNA was isolated from a normal individual and a patient suffering from cancer. The mRNA was allowed to hybridise with the *p53* gene. The schematic diagram shows the results of the hybridisation process under the electron microscope.



Which of the following could be a possible explanation why the patient is suffering from cancer?

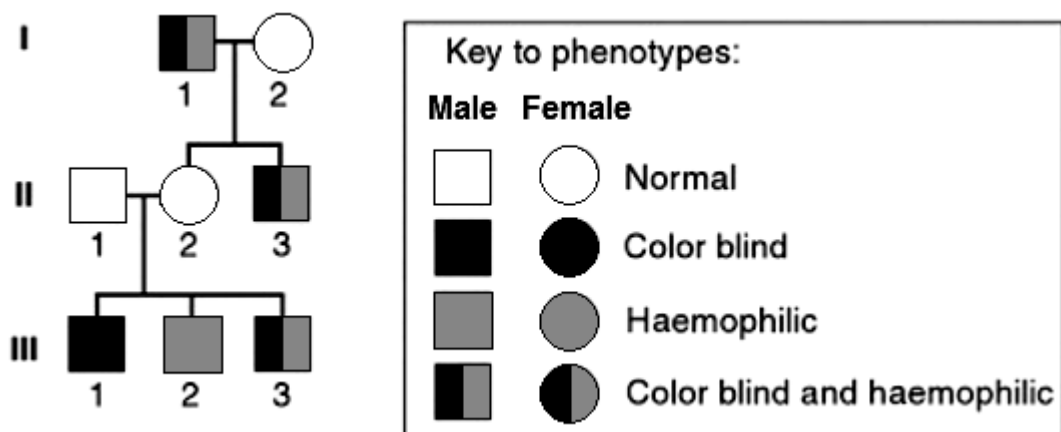
- A A point mutation had occurred in the intron leading to the failure to excise one intron, hence leading to a longer dysfunctional protein being translated.
 - B A point mutation had occurred in the intron leading to an exon being excised, hence leading to a shorter dysfunctional protein being translated.
 - C A point mutation had occurred leading to the failure of spliceosome to recognise splice sites leading to the excision of the wrong intron, leading to a dysfunctional protein being translated.
 - D Gene amplification had occurred leading to the multiple copies of a trinucleotide repeat in an intron, hence causing splice site to be misread due to frameshift mutation, leading to a longer dysfunctional protein being translated.
- 15 A cell, in the midst of actively dividing cells in a *lilium* root tip, was found to be arrested in its cell cycle with an intact nucleus. Which of the following is/are the likely cause(s)?
- 1 Damaged DNA or incomplete replication of DNA.
 - 2 Homologous chromosomes are unable to pair up.
 - 3 Incomplete formation of the mitotic spindle resulting in some chromosomes not attached to fibres.
 - 4 Centrioles fail to replicate.
- A 2, 3 and 4 only
 - B 1, 2 and 3 only
 - C 1 and 2 only
 - D 1 only

16 Which of the following mutations would increase the probability of getting cancer in an individual?

- 1 Gain-of-function mutation in a copy of the p53 gene.
- 2 Somatic mutation in the second copy of the p53 gene.
- 3 Deletion of a single nucleotide at the third position in a copy of the Ras gene.
- 4 Gene amplification of the Ras gene, leading to multiple copies of the gene.

- A** 1, 2, 3 and 4
B 2, 3 and 4 only
C 1 and 3 only
D 2 and 4 only

17 Haemophilia and red-green colour blindness are both sex-linked genetic disorders caused by recessive alleles. The following pedigree chart shows a family's history of the two disorders.



Assuming that the alleles for haemophilia and colour blindness were found on the same chromosome in generation I, which of the following individuals' phenotype is a result of recombination between the haemophilia and colour blindness genes?

- A** II-3 and III-1
B II-3 and III-3
C III-1 and III-2
D III-2 and III-3

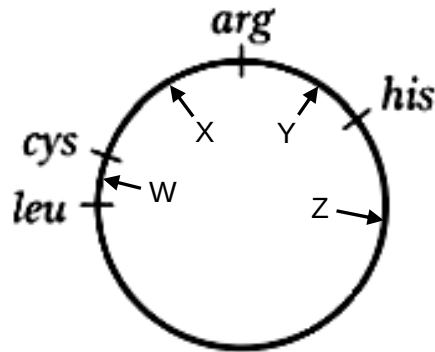
- 18** Three different medical therapies have been used for the treatment of influenza infection. Each therapeutic agent targets a different stage of the viral replicative cycle. The effects of each therapy is described as follows.

Therapeutic agent	Effects
Ribavirin	Competitive inhibitor which is structurally similar to a nucleoside
Peramivir	Inhibitor of viral neuraminidase
Seasonal flu vaccine	Artificially induces active immunity

Which of the following correctly arranges the therapies in order of the stages of the replicative cycle they exert their effects, sorted in order of the earlier stages to the later stages?

- A** Peramivir, ribavirin, seasonal flu vaccine
- B** Ribavirin, peramivir, seasonal flu vaccine
- C** Ribavirin, seasonal flu vaccine, peramivir
- D** Seasonal flu vaccine, ribavirin, peramivir

- 19 *Salmonella typhi* bacteria is known to be a viable host for a newly discovered temperate phage, but the site of prophage integration is unknown. The following gene map shows the loci of four genes on the *S. typhi* chromosome – *arg*, *his*, *leu* and *cys* – responsible for the biosynthesis of four essential amino acids. Four possible prophage integration sites, W, X, Y, Z are indicated.



The phages are allowed to replicate using a strain of *S. typhi* capable of synthesising all four amino acids (*arg*⁺ *his*⁺ *leu*⁺ *cys*⁺), and the replicated phages are then added to a mutant strain of *S. typhi* of genotype *arg*⁻ *his*⁻ *leu*⁻ *cys*⁻.

After a short incubation, samples of these bacteria are plated on four different media supplemented with different amino acids. The following table shows whether colonies were observed on the various media (+ indicates the presence of an amino acid in the medium while – indicates its absence).

Medium	Supplementation of amino acids in medium				Presence of colonies
	Arg	His	Leu	Cys	
1	–	+	+	+	No
2	+	–	+	+	No
3	+	+	–	+	Yes
4	+	+	+	–	Yes

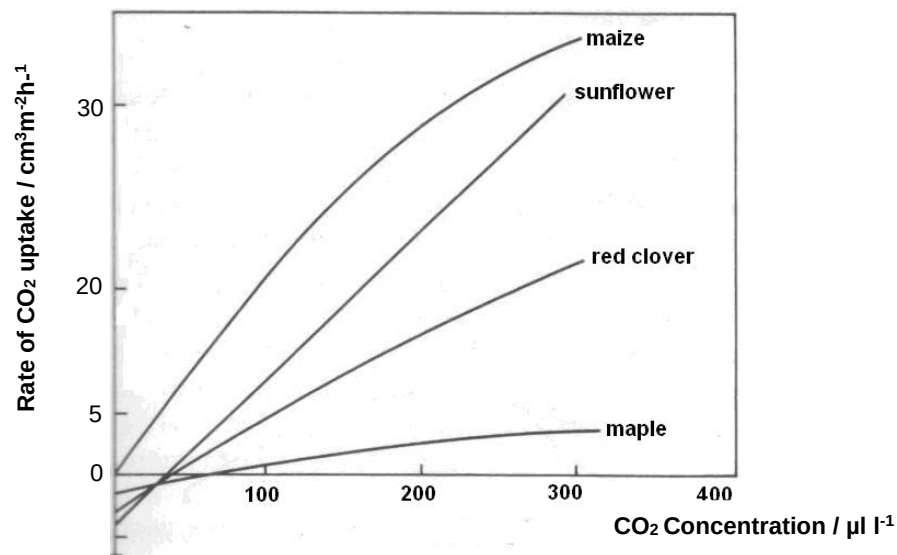
Which of the following is the most likely prophage integration site?

- A Site W
- B Site X
- C Site Y
- D Site Z

20 Which of the following statements explains the mechanism for chemiosmosis in the synthesis of ATP?

- A** The energy released by the reduction and subsequent oxidation of components of the electron transport chain is transferred to ATP synthase for the synthesis of ATP.
- B** Phosphorylation of ADP is linked to the proton gradient established by the electron transport chain.
- C** The difference in pH between the intermembrane space and the cytosol drives the formation of ATP.
- D** The flow of H^+ through ATP synthase into the intermembrane space drives the synthesis of ATP.

21 The diagram below shows the rate of photosynthesis of four different plants at different concentrations of carbon dioxide.

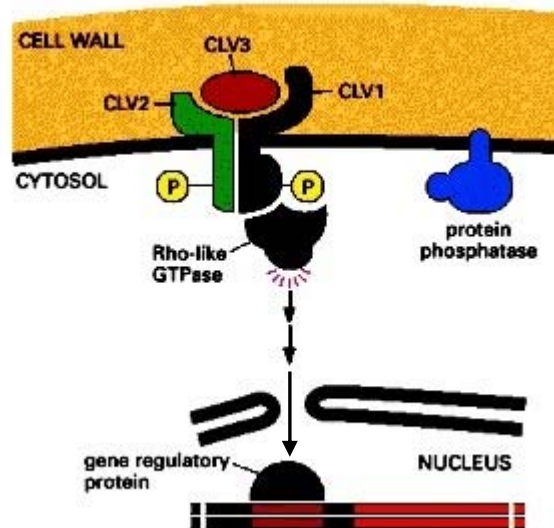


Which of the following conclusions can be made?

- 1 At CO_2 concentrations below $150 \mu l l^{-1}$, CO_2 concentration is the main limiting factor for all the plants.
- 2 CO_2 compensation point is around $40 \mu l l^{-1}$ for sunflower and red clover, and it measures the light intensity when the rate of CO_2 uptake equals to the rate of CO_2 given off.
- 3 Rate of CO_2 uptake was zero for maize at CO_2 concentration of $0 \mu l l^{-1}$ as the amount of CO_2 released from respiration is used for photosynthesis.
- 4 Of the four plants, maple has the lowest amount of organic compound produced at CO_2 concentration of $200 \mu l l^{-1}$.

- A** 1, 2 and 3 only
- B** 1, 3 and 4 only
- C** 1 and 2 only
- D** 3 and 4 only

- 22** Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternative route for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle be affected in chloroplasts poisoned by dinitrophenol?
- A** No change in rate as Calvin cycle occurs in the stroma and not at thylakoid membranes.
 - B** The rate of Calvin cycle will increase as pH in the stroma will decrease towards the optimum for enzymes involved in the cycle.
 - C** The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate.
 - D** The rate of Calvin cycle will decrease with the accumulation of glyceraldehyde-3-phosphate.
- 23** The diagram below shows a cell signalling pathway involved in plant cells.

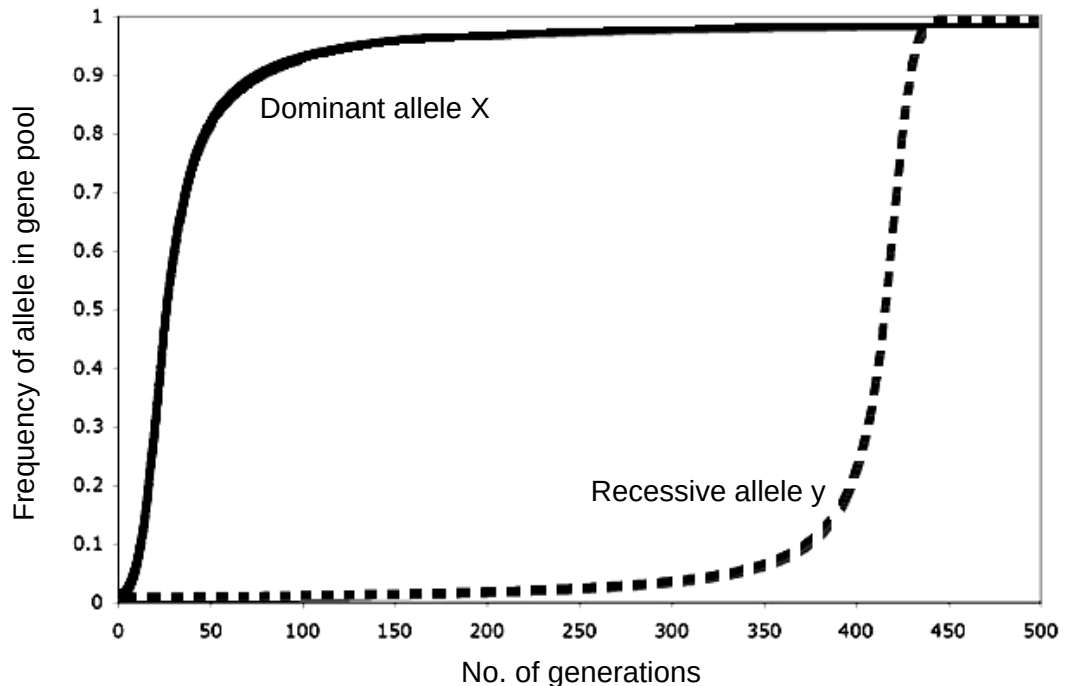


Which of the following sequences regarding the activation of the above signaling pathway is correct?

- A** Dimerization of CLV1 and CLV2 leads to the binding of CLV3 which in turns causes phosphatase to phosphorylate the residues in the cytoplasmic tails, causing the activation of Rho-like GTPase that brings about transcription.
- B** Dimerization of CLV1 and CLV2 leads to activation of CLV3 that causes autophosphorylation of the residues in the cytoplasmic tails which in turn activate Rho-like GTPase that results in DNA replication.
- C** Binding of CLV3 to both CLV1 and CLV2 causes dimerization that leads to autophosphorylation of the residues in the cytoplasmic tails resulting in activation of phosphatase enzyme which in turn phosphorylates proteins needed for protein synthesis.
- D** Binding of CLV3 to both CLV1 and CLV2 causes dimerization that brings about autophosphorylation of the residues in the cytoplasmic tails which in turn activates Rho-like GTPase that results in transcription.

- 24** The mechanisms of natural selection leads to changes in the frequency of alleles in the gene pool of a population over many generations. When a new allele arises due to mutation, the trait coded for by the new allele may be selected for if it confers a selective advantage. The rate at which this new allele approaches fixation – the condition where the allele is found in all individuals of a population – depends on whether it acts in a dominant or recessive manner.

The following graph shows how a dominant allele of gene X (allele X) approaches fixation over time as compared to a recessive allele of another gene Y (allele y). Both alleles confer advantageous traits to the host organism.



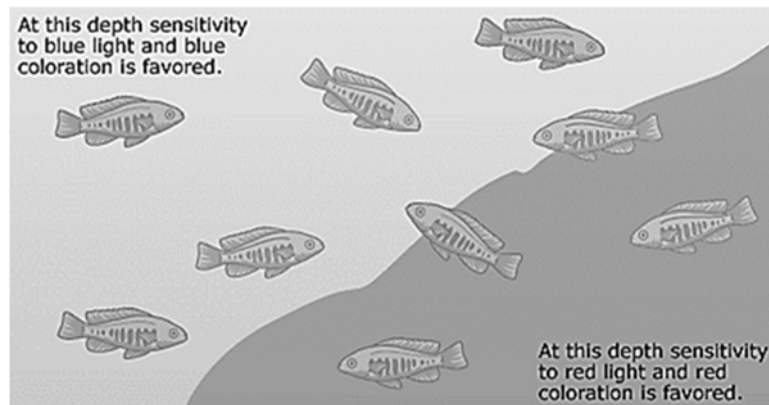
Which of the following explains the differences in rate at which the two types of alleles approach fixation?

- A** The initial increase in frequency of the recessive allele y is slower as heterozygotes have a selective advantage over homozygotes.
- B** The initial increase in frequency of the dominant allele X is faster as the effects of the allele is expressed in the phenotype of heterozygotes.
- C** Fixation of the dominant allele X takes a greater number of generations due to the mechanisms of genetic drift.
- D** Fixation of the recessive allele y takes a smaller number of generations due to mechanisms which preserve heterozygosity in a population.

- 25 The cichlid family of fishes living in Africa's Lake Victoria have undergone extensive adaptive radiation. In the cloudy waters of Lake Victoria, the shallow parts of the lake are dominated by blue light. The depths of the lake, however, are dominated by red light due to the absorption by sediment particles. Cichlids with colour vision sensitive to blue light prefer to dwell in shallow waters of the lake, while cichlids with colour vision sensitive to red light dwell in the deeper waters. Light sensitivity is thought to correlate with the cichlid's ability to survive.

Male cichlids also display a wide variation of body coloration. Males with blue body coloration appear brighter in the shallow waters and have greater reproductive success there, while males with red body coloration appear brighter in the deeper waters and experience greater reproductive success at these depths.

The following figure summarises the observations made about these cichlids.



How many of the following statements regarding the speciation process of cichlids is true?

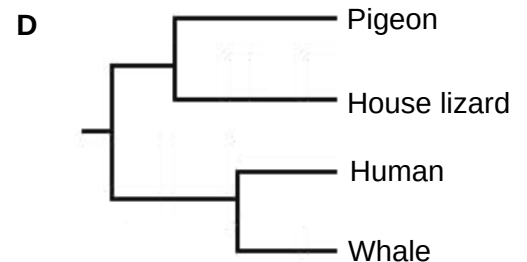
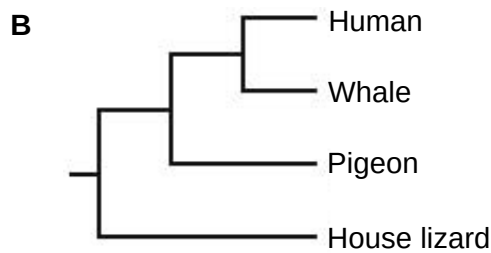
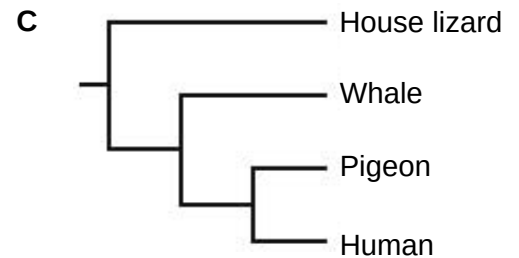
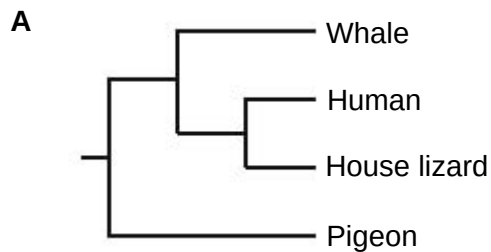
- 1 Disruptive selection has occurred due to variations in light sensitivity in cichlids.
- 2 Speciation is driven by preferential mate selection by female cichlids.
- 3 Allopatric speciation due to geographical isolation has taken place.
- 4 The mechanisms for reproductive isolation in these cichlids are pre-zygotic.

- A** 1
B 2
C 3
D 4

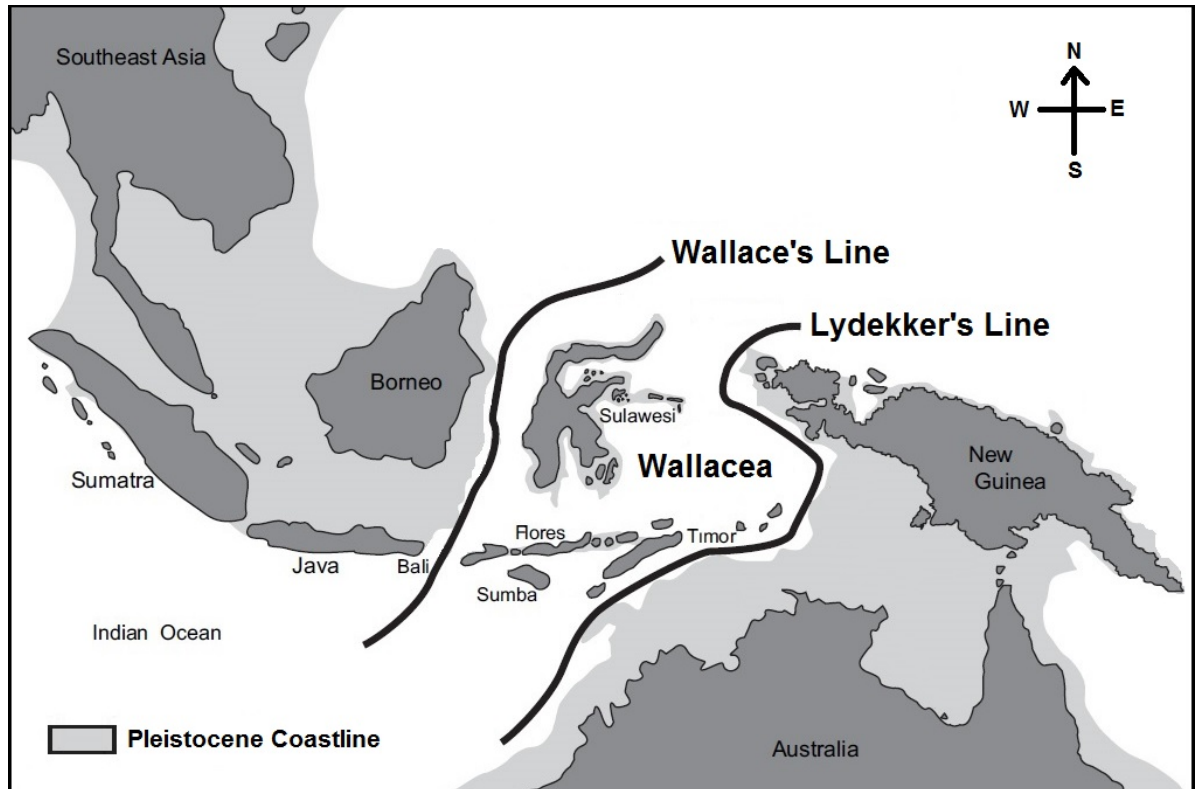
- 26 The phylogenetic relationship of four different species – human, whale, pigeon and the house lizard – was investigated. Part of the amino acid sequence for the cytochrome c protein found in the different species was analysed and is shown in the following table (the letters denote different amino acids).

Species	Amino acid sequence of cytochrome c			
Human	I F V G I K K K E E	R A D L I A Y L K K	A T N E	
Whale	I F A G I K K K G E	R A D L I A Y L K K	A T N E	
Pigeon	I F A G I K K K A E	R A D L I A Y L K Q	A T A K	
House Lizard	I F A G I K K K A E	R A D L I A Y L K D	A T S K	

Which of the following phylogeny of these four species agrees with the available evidence?



- 27 As shown in the following map, in the regions between the islands of Indonesia and Australia, two biogeographical lines were drawn up by European naturalists Alfred Russel Wallace and Richard Lydekker. During the Pleistocene Epoch, lasting from 2 million years till 10,000 years ago, periods of ice ages had lower sea levels. The Pleistocene coastline during ice ages is also indicated in the map.



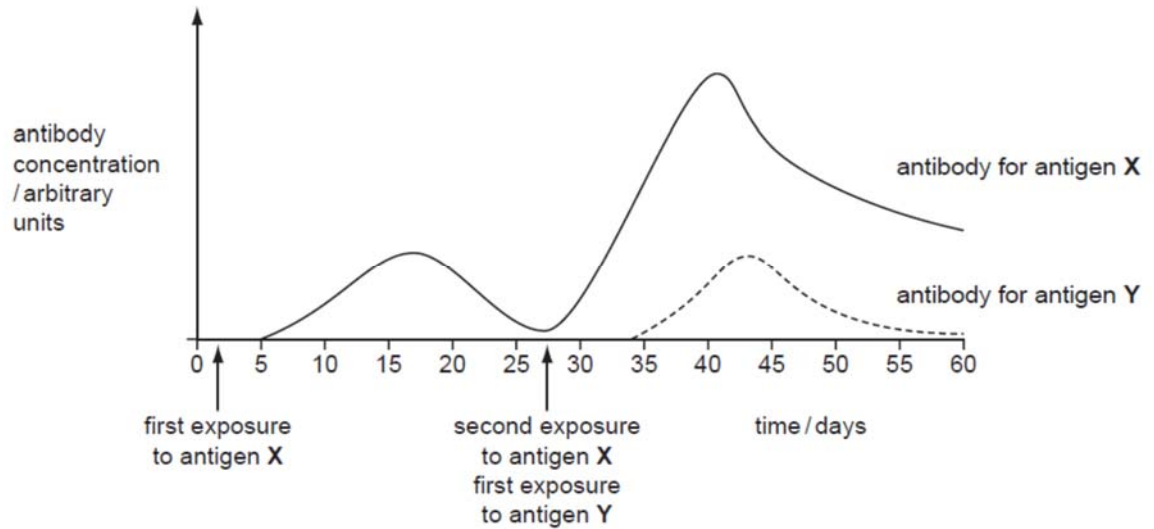
The islands found west of the Wallace's line are rich in biodiversity, represented by many species common with the Southeast Asian mainland such as tigers, rhinoceros, apes and other placental mammals. To the east of Lydekker's line, many marsupials and birds exclusive to Australia populate these regions.

The islands of Wallacea found between the two lines, including Sulawesi and nearby islands, is relatively species-poor. Only some birds, reptiles and insect species of Asian or Australian origin are found there. Flightless birds and freshwater fishes common to Asia or Australia are not found on these islands.

Which of the following statements does **not** explain the geographical distribution of biological species in this region in the present day?

- A Placental mammals from the Southeast Asian mainland were able to populate the islands of Borneo, Sumatra and Java during the Pleistocene ice ages.
- B The islands of Wallacea were geographically isolated from Southeast Asia and Australia even during Pleistocene ice ages due to the presence of deep water bodies.
- C Not all bird species found in this region are able to overcome geographical isolation due to the presence of water bodies.
- D Before the Pleistocene Epoch, the land masses of Southeast Asia and Australia were joined together as a single continent.

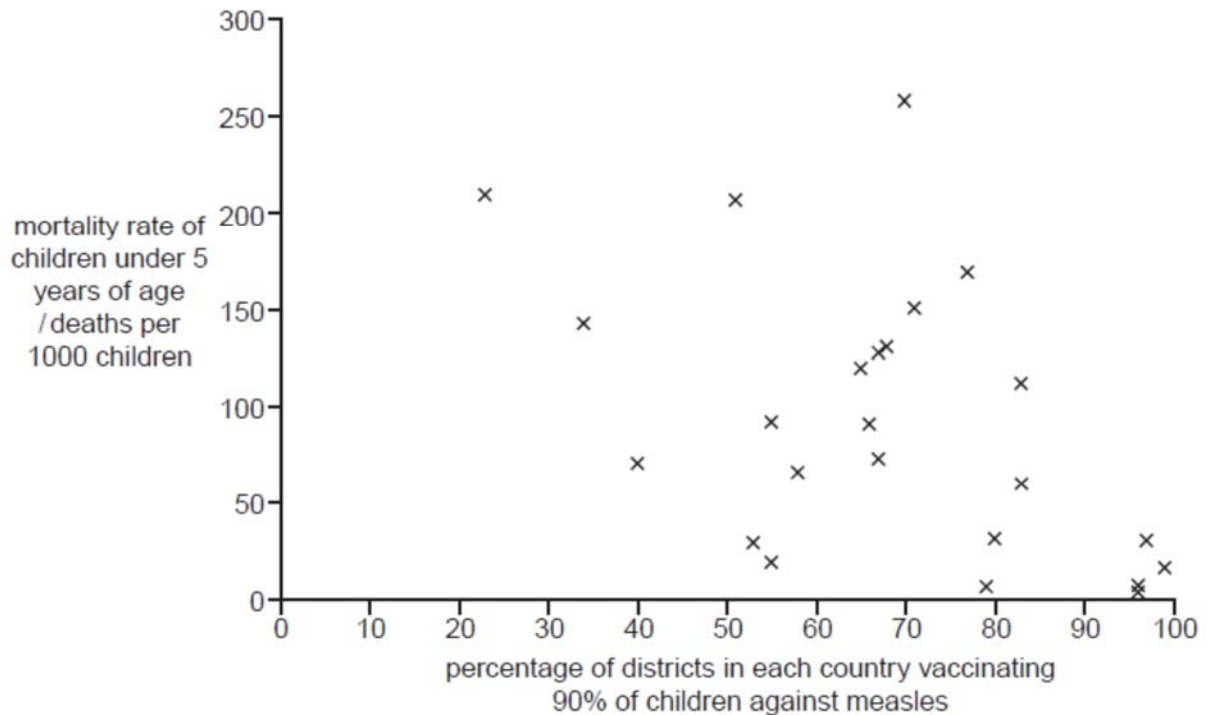
- 28** In an investigation into the immune response, a volunteer was exposed to two different antigens, X and Y. The relative antibody concentration in the blood was measured at regular intervals over 60 days. The graph shows the time when the volunteer was exposed to each antigen and the antibody concentration against time for antigens X and Y.



Which of the following is a valid explanation for the results displayed on the graph?

- A** The innate immune response is activated from day 5, which triggers the activation of the adaptive immunity against antigen X from day 27.
- B** The adaptive immune response against antigen Y is not activated due to the lack of a second exposure to antigen Y at day 60.
- C** Memory B cells specific to antigen X produced by the adaptive immune response enabled a secondary immune response to occur between day 27 and 60.
- D** Memory B cells that remains at the end of day 27 undergo rapid clonal selection to produce antibody against antigen X and Y.

- 29 The World Health Organization (WHO) publishes data on the vaccination programmes for infectious diseases. Each health authority in a country reports its success in vaccinating children in their district. The WHO collects statistics on mortality rates of children under the age of 5 from all causes, including infectious diseases. The figure shows these statistics for 24 countries for the year 2007.



Which of the following can be concluded from the information provided above?

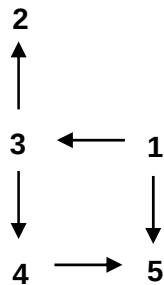
- A Vaccination of 90% of the entire population is recommended as countries with more than 90% of districts reporting 90% of children vaccinated have very low death rates.
- B The mortality rate of children is inversely correlated to the percentage of districts in each country vaccinating 90% of children against measles.
- C Variation between countries with similar percentage of districts reporting 90% of children vaccinated could be attributed to deaths due to infectious diseases.
- D When 90% of children in a district is vaccinated, mortality rate of the children is reduced.

- 30** Investigations into the possible current and future impacts of climate change need to be put into the context of the well-documented and on-going impacts of other drivers of change, such as population growth. The statements below are effects of climate change and population growth.

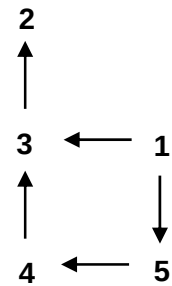
- 1 More intensive land use results in degradation of soils and more rainfall run-off
- 2 Food and feed shortages
- 3 Greater frequency of water deficit in soil for crops and pasture growth
- 4 More frequent dry years experienced
- 5 Increase food demand and competition for pastures

Which of the following diagram correctly illustrates the relationship between climate change and population growth?

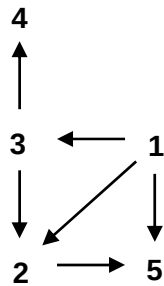
A



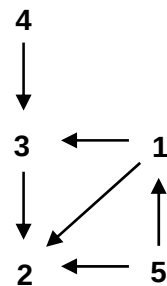
B



C



D



ACJC Prelim 2017
H2 Biology Paper 1 (9744/01) Answers

Question	Answer	Question	Answer
1	B	16	D
2	B	17	C
3	B	18	D
4	B	19	A
5	D	20	B
6	B	21	B
7	C	22	C
8	A	23	D
9	A	24	B
10	D	25	C
11	A	26	D
12	B	27	D
13	A	28	C
14	A	29	B
15	D	30	D